Mukundh Murthy

Seeking a summer internship or shadowing opportunity in which I can grow and leverage my skills, passion and curiosity in an innovative environment.

Work Experience

Researcher

Indian Institute of Technology Madraas / July 2019 – August 2019

I shadowed a PhD and undergraduate student at IITM while also conducting my own mini project to find inhibitors for N-nicotinamide methyltransferase (enzyme involved in cancer).

- Learned to understand catalytic mechanism for hydrolases.
- Learned to synthesize enzymes and conduct fluorescence assays.
- Conducted small molecule docking to receptor using Autodock vina.

Innovator

The Knowledge Society / September 2019 – Present

The Knowledge Society is a 10-month innovation incubator for young, ambitious teenagers interested in science, technology and philosophy.

- Developed a CNN algorithm using Keras to find the free energy of an RNA structure with sequence input.
- Programmed a conditional variational autoencoder using PyTorch to provide minimum nutritional advice to alleviate the greatest number of symptoms.
- Currently developing an LSTM using Keras and novel QSAR algorithm to identify drugs for 2019-nCoV.
- Diving deep into other topics like gene editing, optogenetics, Artificial General Intelligence, Structural Causal Models.
- Have attended multiple conferences including Radcliffe Institute conference on Gene Editing, International Conference on Drug Discovery, and will be attending World Summit AI in March.

Consultant

Kidogo / March 2020 – Present

I am currently consulting for Kidogo, a social enterprise providing early childhood education to low-income communities in East Africa. My team and I at The Knowledge Society are working on creative ways to increase the income for caregivers.

Consultant

Techstars Boston / November – December 2019

I consulted for Techstars Boston, a three-month mentorship-driven startup accelerator in more than 100 countries. My team and I provided Techstars Boston with a recommendation on how to increase the inbound of mentors using natural language processing.

Media and Public Speaking Experience

- Speaker at Global Silicon Valley Labs at an event affiliated with The Knowledge Society and Harvard Alumni for Education. I spoke about my work in RNA bioinformatics and the importance of understanding the function of non-coding elements of our genome.
- Potential speaker at Rework AI in Healthcare conference (Boston)

EDUCATION

Acton-Boxborough Regional High School (2017-2021)

Current Honors student with 4.0 GPA, in addition to taking online classes in math and science.

Medicinal Chemistry (2019) – DavidsonX

Intro to Deep Learning with Pytoch (2019) – Udacity

Genomic Data Science with Galaxy (2019) - Coursera

Probabilistic Systems Analysis and Applied Probability (2020) – MIT Open Courseware

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SKILLS

Professional

Creative problem-solver
Keen listener
Flexible and adaptable team player
Eager to learn

Technical

Wet lab experience
Experimental design
Biochemistry
Statistical Analysis
Autodock
PyMol
PyTorch
TensorFlow
Microsoft Word
Microsoft Excel

INTERESTS

Drug Development
Artificial Intelligence
Causal Statistics
Gene editing
Genomics
Proteomics

Awards and Accomplishments

2016 - 2nd Place in You Be the Chemist State Competition

2019 – 2nd Place in Environmental Chemistry at the National Science Olympiad Competition

2019 – 1st Place in Chemistry and Protein Modeling at the Harvard Invitational and Massachusetts State Science Olympiad Tournaments

2020 – Biobuilders (high school synthetic biology competition) – Paper on splinter degradation won award for problem solving

2020 – 2nd Place at MIT's Latin America AI Hackathon - competition to solve problems in crime, climate change, healthcare, or education in Latin America using AI

2021 – National Merit Scholarship Qualifier (perfect PSAT: 1520)